

Smart CareerPilot Platform for Intelligent Career Design

Project II (CS3202)

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Introduction & Problem Statement

Introduction

- All-in-one career development platform for CSE students.
- Integrates AI/ML for personalized guidance, resume analysis, job search, and interview prep.
- Bridges academic learning with employability using modern tech.

Problem Statement

- Fragmented career tools lack integration and personalization.
- Job seekers face challenges in resume optimization, job discovery, and interview readiness.
- Need for a unified, AI-driven platform to streamline career advancement.

Objectives

- Develop a comprehensive career platform with AI personalization.
- Streamline job search and application processes.
- Enhance interview prep with analytics and skill gap analysis.

Background and Existing Work

Existing Solutions

- **LinkedIn, Indeed:** Job listings without personalized guidance[4] .
- **Zety, Resumonk:** Resume builders without AI optimization [7].
- **Jobscan:** Partial resume analysis; lacks full career support [1].

Limitations

- Fragmented tools, no unified workflow.
- No AI-based personalization or skill tracking.
- Missing integration of resume, job search, interview, and learning.

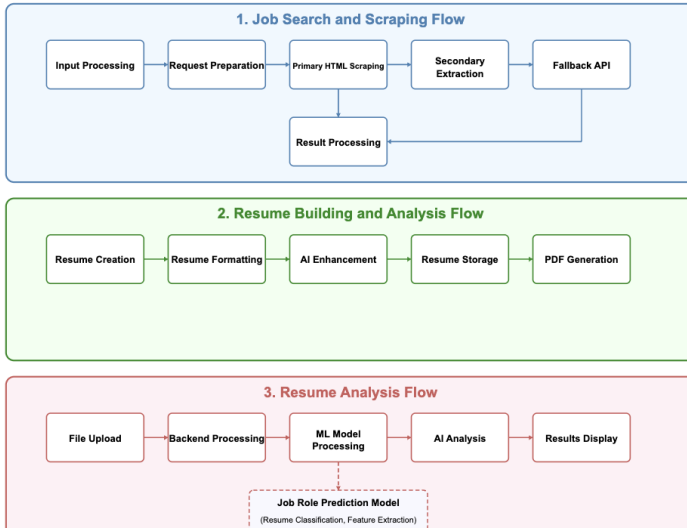
Why *CareerPilot*?

- Unified platform tailored for students.
- AI/ML-powered resume checks, job discovery, interview prep.
- Tracks weaknesses, suggests personalized improvements.

System Design (Part 1)

Process Flow Diagram

CareerPilot System Process Flow



System Design (Part 2)

Process Flow Diagram(contd.)

4. Cover Letter Generation Flow



5. Interview Preparation Flow



6. Todo List Management



7. User Authentication Flow



8. Industry Insights Flow



Technical Skills & Implementation

Programming & Frameworks

- **Languages:** JavaScript, Python
- **Frontend:** Next.js (React), TailwindCSS, Shadcn UI[5]
- **Backend:** FastAPI
- **AI/ML:** Gemini API [2], Scikit-learn (XGBoost, LightGBM, Random Forest, BERT)

Data & Tools

- **Database:** PostgreSQL, Prisma ORM [6]
- **Web Scraping:** Axios, Cheerio
- **Tools:** VS Code, DevTools, Git, GitHub, Puppeteer, Clerk
- **Concepts:** REST APIs, JSON, Agile, TF-IDF, feature engineering

Implementation Overview

- **Resume Builder:** AI analysis (87%), ATS optimization, PDF export
- **Job Search:** LinkedIn/Indeed scraping (95%), filtering
- **Interview Prep:** Mock interviews, DSA practice, analytics
- **Career Guidance:** Skill gap analysis, learning paths

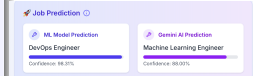
AI/ML Integration

- **Resume Classification:** BERT [3], 87% accuracy (25K data)
- **Job Matching:** XGBoost/LightGBM, 100% test accuracy
- **Gemini AI:** Feedback, cover letters, interview analysis

Results

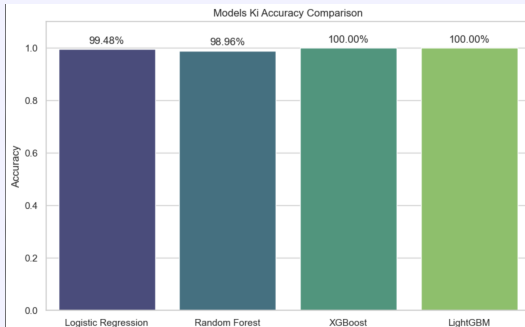
Performance Metrics

Feature	Metric
Resume Analysis	87% accuracy, 15s
Job Search	95% retrieval, 3s
Interview Prep	4.3/5 user rating
Usability	4.2/5, 92% task completion



ML-Based Job Role Prediction

Model Performance



Conclusion & Future Work

Key Achievements

- Unified AI-powered career platform for CSE students.
- High performance: 87% resume accuracy, 95% job retrieval.
- Intuitive UI with 4.2/5 usability score.
- Democratizes career resources for diverse users.

Future Enhancements

- AI chatbot for real-time assistance and interview monitoring via computer vision.
- PWA for offline access with multilingual support.
- Integration with more job platforms (e.g., Glassdoor) to expand opportunities.



J. Alvarez and R. Singh.

Learning to rank resumes for intelligent candidate screening, 2023.



Google DeepMind.

Gemini api, 2024.



Jacob Devlin et al.

Bert: Pre-training of deep bidirectional transformers for language understanding, 2019.



LinkedIn.

Linkedin: Professional network, 2025.



Shadcn.

shadcn/ui, 2024.



Prisma Team.

Prisma orm, 2024.



Zety.

Resume building made easy, 2025.